Institute of Technology of Cambodia

Java Report Course:

Object Oriented Programming

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Example 1

```cpp
#include <iostream>
using namespace std;
int main(){
    int age;
    string name, gender, major;
    cout << "Please enter your name: ";
    cin >> name;
    cout << "Please enter your major: ";
    cin >> major;
    cout << "Please enter your gender (male or female): ";
    cin >> gender;
    cout << "Please enter your age: ";
    cin >> age;
    if ( gender.compare("male")==0){
        cout << "Hi, Mr "<<name<<"! your age is "<<age<<" year old and you learn "<<major<<endl;
    }
    if ( gender.compare("female")==0){
        cout << "Hi, Ms "<<name<<"! your age is "<<age<<" year old and you learn "<<major<<endl;
    }
    if (age>=18){
        cout<<"you can vote!";
    }
    else{
        cout<<"you cant vote!";
    }
    return 0;
}
```

```
Please enter your name: both
Please enter your major: se
Please enter your gender (male or female): male
Please enter your age: 18
Hi, Mr both! your age is 18 year old and you learn se
you can vote!
```

Example 2

```cpp
#include <iostream>
using namespace std;
int main(){
    cout<<"Enter a character: ";
    char ch;
    cin>>ch;
    if (string("aeiouAEIOU").find(ch) != string::npos){
        cout<<"Vowel"<<endl;
    }
    else if (isalpha(ch)){
        cout << "Consonant" <<endl;
    }
    return 0;
}
```

```
Enter a character: s
Consonant
```

Example 3

```
#include <iostream>
using namespace std;
int main(){
    string name, gender;
    double salary;
    cout<<"Enter name : ";
    cin>>name;
    cout<<"Enter gender (male or female): ";
    cin>>gender;
    cout<<"Enter salary : ";
    cin>>salary;
    double tax;
    if (gender == "male"){
        if (salary > 1000){
            tax = salary*0.095;
        }
        if (salary <=1000 && salary >500){
            tax = salary*0.07;
        }
        if (salary<=500 && salary>300){
            tax = salary*0.05;
        }
    }
    else if (gender == "female"){
        if (salary > 1000){
            tax = salary*0.08;
        }
        if (salary <=1000 && salary >500){
            tax = salary*0.065;
        }
        if (salary<=500 && salary>300){
            tax = salary*0.035;
        }
    }
    else {
        tax = 0;
    }
    cout<<"Your tax are : "<<tax;
    return 0;
}
```

```
Enter name : both
Enter gender (male or female): male
Enter salary : 30000
Your tax are : 2850
```

Example 4

```
#include <iostream>
using namespace std;

int random(int min, int max){
    return min + rand() % (max-min+1);
}
int main(){
    // Create a C++ program that can play a game Rock-Paper-Scissor.
    // The game plays between the user against the computer.
    srand(time(0));
    int user_score = 0, computer_score = 0;
    string option[] = {"rock","paper","scissor"};
    int i;

    while(true) {
        cout<<"What will the user choose?"<<endl;
        cout<<"1. Scissor"<<endl;
        cout<<"2. Paper"<<endl;
        cout<<"3. Rock"<<endl;
        cout<<"4. Exit"<<endl;
        cout<<"option : ";
        cin>>i;
        cout<<endl<<endl;

        if(i == 4) {
            cout<<"Exiting the game..."<<endl;
            break;
        }

        if(i >= 1 && i <= 3) {
            int j = random(0,2);
            string opp = option[j];
            cout<<"User picks : "<< option[i-1]<<"\tI pick :"<<opp<<endl;
            if(option[i-1] == "rock" && opp == "scissor") {
                cout<<"User wins!"<<endl;
                user_score++;
            }
            else if(option[i-1] == "paper" && opp == "rock") {
                cout<<"User wins!"<<endl;
                user_score++;
            }
            else if(option[i-1] == "scissor" && opp == "paper") {
                cout<<"User wins!"<<endl;
```

```
                user_score++;
            }
            else if(option[i-1] == opp) {
                cout<<"It's a tie!"<<endl;
            }
            else {
                cout<<"Computer wins!"<<endl;
                computer_score++;
            }
            cout<<"Score - User: "<<user_score<<" | Computer:
"<<computer_score<<endl<<endl;
        }
        else {
            cout<<"Invalid option!"<<endl<<endl;
        }
    }

    cout<<"\nFinal Score - User: "<<user_score<<" | Computer:
"<<computer_score<<endl;
    return 0;
}
```

```
What will the user choose?
1. Scissor
2. Paper
3. Rock
4. Exit
option : 3


User picks : scissor    I pick :rock
Computer wins!
Score - User: 0 | Computer: 1

What will the user choose?
1. Scissor
2. Paper
3. Rock
4. Exit
option :
```

Example 5

```cpp
#include <iostream>
using namespace std;

int main() {
    long long minutes;

    cout << "Enter minute(s): ";
    cin >> minutes;

    if (minutes < 0) {
        cout << "Minutes must be non-negative." << endl;
        return 1;
    }

    long long hours = minutes / 60;
    long long remainingMinutes = minutes % 60;
    long long seconds = remainingMinutes * 60;

    cout << "Time format (h:m:s) = "
         << hours << ":" << remainingMinutes << ":" << seconds << endl;

    return 0;
}
```

```
Enter minute(s): 300
Time format (h:m:s) = 5:0:0
```

Example 6

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;

    cout << "Enter n (n > 50): ";
    cin >> n;

    while (n <= 50) {
        cout << "Invalid input. Please enter n greater than 50: ";
        cin >> n;
    }

    long long sum = 0;
    for (int i = 1; i <= n; i++) {
        if (i == 10 || i == 30) {
            continue;
        }
        sum += i;
    }

    cout << "Summation from 1 to " << n
         << " (except 10 and 30) = " << sum << endl;

    return 0;
}
```

```
Enter n (n > 50): 234
Summation from 1 to 234 (except 10 and 30) = 27455
```

Example 7

```cpp
#include <iostream>
#include <iomanip>
#include <string>
#include <vector>
using namespace std;

struct Book {
    int bookID;
    string isbn;
    string title;
    int publishedYear;
    vector<string> authorNames;
    double price;
};

void displayBookByISBN(Book books[], int size, string isbn) {
    for (int i = 0; i < size; i++) {
        if (books[i].isbn == isbn) {
            cout << "Book found:" << endl;
            cout << "Book ID: " << books[i].bookID << endl;
            cout << "ISBN: " << books[i].isbn << endl;
            cout << "Title: " << books[i].title << endl;
            cout << "Published Year: " << books[i].publishedYear << endl;
            cout << "Authors: ";
            for (size_t j = 0; j < books[i].authorNames.size(); j++) {
                cout << books[i].authorNames[j];
                if (j + 1 < books[i].authorNames.size()) {
                    cout << ", ";
                }
            }
            cout << endl;
            cout << fixed << setprecision(2);
            cout << "Price ($): " << books[i].price << endl;
            return;
        }
    }

    cout << "No book with ISBN " << isbn << " was found." << endl;
}

void displayAllBooks(Book books[], int size) {
    cout << "\nAll books:" << endl;
    for (int i = 0; i < size; i++) {
```

```cpp
        cout << "------------------------------" << endl;
        cout << "Book ID: " << books[i].bookID << endl;
        cout << "ISBN: " << books[i].isbn << endl;
        cout << "Title: " << books[i].title << endl;
        cout << "Published Year: " << books[i].publishedYear << endl;
        cout << "Authors: ";
        for (size_t j = 0; j < books[i].authorNames.size(); j++) {
            cout << books[i].authorNames[j];
            if (j + 1 < books[i].authorNames.size()) {
                cout << ", ";
            }
        }
        cout << endl;
        cout << fixed << setprecision(2);
        cout << "Price ($): " << books[i].price << endl;
    }
    cout << "------------------------------" << endl;
}

int main() {
    Book books[5] = {
        {1, "1234", "The C Programming Language", 1988, {"Brian W. Kernighan",
"Dennis M. Ritchie"}, 45.50},
        {2, "7654", "Effective C++", 2005, {"Scott Meyers"}, 39.99},
        {3, "3545", "Design Patterns", 1994, {"Erich Gamma", "Richard Helm",
"Ralph Johnson", "John Vlissides"}, 55.75},
        {4, "9999", "Clean Code", 2008, {"Robert C. Martin"}, 42.00},
        {5, "8888", "Fluent Python", 2015, {"Luciano Ramalho"}, 49.90}
    };

    displayAllBooks(books, 5);

    string searchISBN;
    cout << "\nEnter ISBN to search: ";
    cin >> searchISBN;

    displayBookByISBN(books, 5, searchISBN);

    return 0;
}
```

```
------------------------------
Book ID: 3
ISBN: 3545
Title: Design Patterns
Published Year: 1994
Authors: Erich Gamma, Richard Helm, Ralph Johnson, Jo
Price ($): 55.75
------------------------------
Book ID: 4
ISBN: 9999
Title: Clean Code
Published Year: 2008
Authors: Robert C. Martin
Price ($): 42.00
------------------------------
Book ID: 5
ISBN: 8888
Title: Fluent Python
Published Year: 2015
Authors: Luciano Ramalho
Price ($): 49.90
------------------------------


Enter ISBN to search: 8888
Book found:
Book ID: 5
ISBN: 8888
Title: Fluent Python
Published Year: 2015
Authors: Luciano Ramalho
Price ($): 49.90
```

Example 8

```
#include <iostream>
#include <cmath>
#include <limits>
using namespace std;

double celsiusToFahrenheit(double celsius) {
    return (celsius * 9.0 / 5.0) + 32;
}

double fahrenheitToCelsius(double fahrenheit) {
    return (fahrenheit - 32) * 5.0 / 9.0;
}

void quadraticRoots(double a, double b, double c) {
    if (a == 0) {
        cout << "Not a quadratic equation because a = 0." << endl;
        return;
    }

    double discriminant = b * b - 4 * a * c;

    if (discriminant > 0) {
        double x1 = (-b + sqrt(discriminant)) / (2 * a);
        double x2 = (-b - sqrt(discriminant)) / (2 * a);
        cout << "Two real roots: x1 = " << x1 << ", x2 = " << x2 << endl;
    } else if (discriminant == 0) {
        double x = -b / (2 * a);
        cout << "One real root: x = " << x << endl;
    } else {
        double realPart = -b / (2 * a);
        double imaginaryPart = sqrt(-discriminant) / (2 * a);
        cout << "Two complex roots: x1 = " << realPart << " + " << imaginaryPart
<< "i"
             << ", x2 = " << realPart << " - " << imaginaryPart << "i" << endl;
    }
}

double computeBMI(double weightKg, double heightM) {
    return weightKg / (heightM * heightM);
}

void showBMICategory(double bmi) {
    if (bmi < 18.5) {
```

```cpp
        cout << "BMI = " << bmi << " -> underweight" << endl;
    } else if (bmi < 25) {
        cout << "BMI = " << bmi << " -> normal weight" << endl;
    } else if (bmi < 30) {
        cout << "BMI = " << bmi << " -> overweight" << endl;
    } else {
        cout << "BMI = " << bmi << " -> obese" << endl;
    }
}

long long sumExceptDivisibleBy3(int n) {
    long long sum = 0;
    for (int i = 1; i <= n; i++) {
        if (i % 3 != 0) {
            sum += i;
        }
    }
    return sum;
}

int main() {
    int choice;

    do {
        cout << "\n===== MENU =====" << endl;
        cout << "1. Celsius to Fahrenheit" << endl;
        cout << "2. Fahrenheit to Celsius" << endl;
        cout << "3. Root of quadratic equation" << endl;
        cout << "4. Compute BMI and category" << endl;
        cout << "5. Sum 1..n except numbers divisible by 3" << endl;
        cout << "0. Exit" << endl;
        cout << "Choose an option: ";
        cin >> choice;

        if (cin.fail()) {
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
            cout << "Invalid input." << endl;
            continue;
        }

        if (choice == 1) {
            double c;
            cout << "Enter Celsius: ";
            cin >> c;
```

```
            cout << "Fahrenheit = " << celsiusToFahrenheit(c) << endl;
        } else if (choice == 2) {
            double f;
            cout << "Enter Fahrenheit: ";
            cin >> f;
            cout << "Celsius = " << fahrenheitToCelsius(f) << endl;
        } else if (choice == 3) {
            double a, b, c;
            cout << "For ax^2 + bx + c = 0" << endl;
            cout << "Enter a, b, c: ";
            cin >> a >> b >> c;
            quadraticRoots(a, b, c);
        } else if (choice == 4) {
            double weight, height;
            cout << "Enter weight (kg): ";
            cin >> weight;
            cout << "Enter height (m): ";
            cin >> height;

            if (height <= 0) {
                cout << "Height must be greater than 0." << endl;
            } else {
                double bmi = computeBMI(weight, height);
                showBMICategory(bmi);
            }
        } else if (choice == 5) {
            int n;
            cout << "Enter n: ";
            cin >> n;
            if (n < 1) {
                cout << "n must be at least 1." << endl;
            } else {
                cout << "Sum = " << sumExceptDivisibleBy3(n) << endl;
            }
        } else if (choice == 0) {
            cout << "Program ended." << endl;
        } else {
            cout << "Invalid choice." << endl;
        }

    } while (choice != 0);

    return 0;
}
```

```
===== MENU =====
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Root of quadratic equation
4. Compute BMI and category
5. Sum 1..n except numbers divisible by 3
0. Exit
Choose an option: 4
Enter weight (kg): 65
Enter height (m): 1.7
BMI = 22.4913 -> normal weight

===== MENU =====
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Root of quadratic equation
4. Compute BMI and category
5. Sum 1..n except numbers divisible by 3
0. Exit
Choose an option: 1
Enter Celsius: 30All books:
Fahrenheit = 86

===== MENU =====
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
3. Root of quadratic equation
4. Compute BMI and category
5. Sum 1..n except numbers divisible by 3
0. Exit
```